

Smart contract audit

rain.lib.typecast

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Project description

This repository contains two Solidity libraries, LibCast and LibConvert, designed for high-performance, gas-optimized type manipulation. LibCast provides functionality to retype array memory pointers (e.g., casting `uint256[]` to `address[]`) without copying data, leveraging shared memory structures for zero-overhead casting. LibConvert handles more complex conversions that involve structural changes, such as in-place modification of array lengths or unsafe packing of data (e.g., truncating `uint256` arrays to packed bytes). These tools are intended for use cases where standard Solidity type safety is traded for execution efficiency and direct memory control.

Executive summary

Type	Library
Languages	Solidity
Methods	Architecture Review, Manual Review, Unit Testing, Functional Testing, Automated Review
Documentation	README.md
Repositories	https://github.com/rainlanguage/rain.lib.typecast

Reviews

Date	Repository	Commit
26/01/26	rain.lib.typecast	fe4b2a4c05632e394b0643990d6de39658fb8fb5

Scope

Contracts
src/LibCast.sol
src/LibConvert.sol



Technical analysis and findings

Critical	0
High	0
Medium	0
Low	0
Informational	0

Security findings

**** Critical

No critical severity issue found.

*** High

No high severity issue found.

** Medium

No medium severity issue found.

* Low

No low severity issue found.

Informational

No informational severity issue found.



General Risks

- **Memory Aliasing & Corruption:** Functions such as `unsafeToBytes` create multiple pointers to the same memory location (aliasing). Modifying the data via one pointer (e.g., as bytes) allows for reading/writing incompatible data structures via the other (e.g., as `uint256[]`), potentially corrupting the memory layout and crashing the execution context.
- **Silent Data Loss (Truncation):** The `LibConvert` library performs operations like strict truncation (e.g., `unsafeTo16BitBytes`) without checking for integer overflow. If the input values exceed the target bit width, the high-order bits are silently discarded, leading to data corruption without reverting the transaction.
- **Bypassing Type Validation:** `LibCast` relies on the caller to ensure data validity. Casting arbitrary integer data to types like function pointers or address arrays does not validate the underlying values. Using invalid values in the new type context may lead to unexpected behavior in downstream logic.



Approach and methodology

To establish a uniform evaluation, we define the following terminology in accordance with the OWASP Risk Rating

Methodology:

	Likelihood Indicates the probability of a specific vulnerability being discovered and exploited in real-world scenarios
	Impact Measures the technical loss and business repercussions resulting from a successful attack
	Severity Reflects the comprehensive magnitude of the risk, combining both the probability of occurrence (likelihood) and the extent of potential consequences (impact)

Likelihood and impact are divided into three levels: High H, Medium M, and Low L. The severity of a risk is a blend of these two factors, leading to its classification into one of four tiers: Critical, High, Medium, or Low.

When we identify an issue, our approach may include deploying contracts on our private testnet for validation through testing. Where necessary, we might also create a Proof of Concept PoC to demonstrate potential exploitability. In particular, we perform the audit according to the following procedure:

	Advanced DeFi Scrutiny We further review business logics, examine system operations, and place DeFi-related aspects under scrutiny to uncover possible pitfalls and/or bugs
	Semantic Consistency Checks We then manually check the logic of implemented smart contracts and compare with the description in the white paper.
	Security Analysis The process begins with a comprehensive examination of the system to gain a deep understanding of its internal mechanisms, identifying any irregularities and potential weak spots.